

AT89C52-BASED ROBOCAR

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A robot can be defined as an electro-mechanical system with the capability of sensing its environment, manipulating it and acting according to the pre-programmed sequence. It is a machine that appears intelligent due to the instructions it receives from a compu-

is driven by an L293D motor driver IC. L293D is a quadruple half-H driver IC with an output current rating of 600mA at voltages ranging from 4.5V to 36V.

A DC motor is used for moving the car forward and backward, depending on the signals received from the three IR sensors. These sensors are used in any remotely-operated home appliance like TV, DVD player, etc. The IR sen-

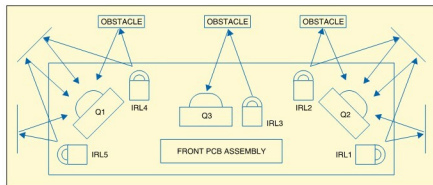


Fig. 1: Arrangement of IR LEDs and sensors on the front PCB

ter inside it which handles multiple tasks. This article features a car robot—RoboCar—which uses a microcontroller to detect obstacles and manipulate its direction as per the inputs from three infra-red (IR) sensors mounted in front of the car.

Basic components of RoboCar

The heart of the system is a microcontroller—Atmel AT89C52. It is programmed to accept inputs from its port p0 to sense the obstacles around it and control the steering to avoid any collision.

There are three TSOP1738 IR sensors (Q1, Q2 and Q3) used in this project—one at the centre and the remaining two on the left and right to detect obstructions, if any, in front of the RoboCar (Fig. 1). In case of an obstacle, or a potential collision, the microcontroller controls the steering through a bipolar stepper motor which

sor TSOP1738 operates at a frequency of 38 kHz. NE555 timer IC is used for generating a pulse of 38 kHz and transmitted through IR LEDs. There are five IR LEDs—two each on the left and right and one on the front of the robot. The reflected IR that beams from the obstacles are received by the sensors and sent to the microcontroller. The microcontroller is programmed in such a way that it takes the decision and changes the path of the robot as per the sensors' inputs to avoid the obstacles.

Circuit description

The schematic diagram of an AT89C52-based RoboCar is shown in Fig. 2. This RoboCar is powered by a 12V rechargeable battery connected to a 5V regulator IC 7805 (IC1) through a current limiter R1. This 5V is used for supplying power to the microcontroller AT89C52, IC NE555 and IR sensors TSOP1738.



Switch S1 is used for turning on the circuit to run the robot. An additional filter circuit, comprising a resistor and a capacitor for each IR sensor, is used for preventing interference with noise signals. Switch S2 acts as a hardware reset for the microcontroller in case the robot is not running properly.

The microcontroller AT89C52 (IC4) is responsible for taking decisions if

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2, IC3	- L293D motor driver
IC4	- AT89C52 microcontroller
IC5	- NE555 timer
T1	- 2N2222 npn transistor
IRL1-IRL5	- 5mm IR LED
LED1	- 5mm red LED
Q1-Q3	- TSOP1738 IR sensor

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

R1	- 3.9-ohms, 2W
R2	- 10-kilo-ohms
R3, R4	- 3.3-kilo-ohms
R5	- 470-ohms
R6-R10	- 22-ohms
R11-R13	- 100-ohms
R14-R21	- 82-kilo-ohms or resistor network
R22	- 1-kilo-ohms
VR1	- 5.1-kilo-ohms preset

Capacitors:

C1, C10	- 0.1 μ F/25V electrolytic
C2	- 10 μ F/25V electrolytic
C3, C4	- 33pF ceramic
C7-C9	- 4.7 μ F/25V electrolytic

Miscellaneous:

BATT	- 4Vx3 rechargeable battery
X _{CRY}	- 14MHz crystal
S1	- SPDT on/off switch
S2	- Tactile switch
M1	- 12V DC motor
M2	- 12V bipolar stepper motor, 7.5°/step
	- Four wheels each of 75mm dia
	- Chassis 265x150mm
	- Heat sink for 7805
	- Pulley and belt
	- Connectors and wires